EREN CETIN

MSc Student at ETH Zurich
Excellence Scholar

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EDUCATION

ETH Zurich *Zurich, Switzerland*
MSc in Electrical Engineering and Information Technology *09/2022 - present*

- Specialization in Machine Learning and Signal Processing
- Awarded with Excellence Scholarship by ETH Foundation
- GPA: 5.82 / 6.00 *(Thesis grade: 6.00 / 6.00, Course average: 5.76/6.00)*
- Related Coursework: Deep Learning, Machine Perception, Computer Vision, Probabilistic AI, Robot Learning (3D Reconstruction, INRs), Large Language Models

Koç University *Istanbul, Turkey*
BSc in Electrical & Electronics Engineering / Double Major in Economics *09/2017 - 06/2022*

- Salutatorian of the class of 2022, Top ranked student in both EEE and Economics
- Awarded with full-tuition scholarship and Vehbi Koç Honor Award each semester
- GPA: 4.00/4.00
- Specialization in Digital Image/Video Processing, Machine Learning, Deep Learning, AI

RESEARCH EXPERIENCE

Disney Research Studios & ETH Zurich *Zurich, Switzerland*
Master's Thesis Student *02/2025 - 09/2025*

- *''[Name Redacted for Review Purposes]''*. Research on utilizing video diffusion models in conjunction with implicit neural representations to achieve low-rate compression.
- Paper submission is on review for *CVPR 2026*. Our work achieves high perceptual quality compared to distortion oriented codecs, leveraging conditional video diffusion models.
- Patent pending on the proposed video diffusion-based video compression framework.
- Python, PyTorch

Computer Vision and Geometry Group, ETH Zurich *Zurich, Switzerland*
Graduate Research Student *06/2024 - 01/2025*

- Research on generalizable sparse view 3D Gaussian splatting for 3D reconstruction. I am working on multi-view stereo approaches with feed-forward neural networks for 3D Gaussian estimation.
- Python, PyTorch, CUDA, C/C++

Computer Vision Lab, ETH Zurich *Zurich, Switzerland*
Graduate Research Student *03/2024 - 06/2024*

- Representing a scene with 3D Gaussian splatting has high storage complexity. This project focused on adoption of learned image compression methods for Gaussian splatting compression.
- Python, PyTorch, CUDA, C/C++

KUIS AI Lab, Koç University *Istanbul, Turkey*
Undergraduate Research Assistant *02/2021 - 09/2022*

- Research on image/video compression using recurrent and fully convolutional auto-encoders under supervision of Murat Tekalp. Presented our work on bidirectional video compression in ICIP 2022.
- Python, PyTorch

MA-CSSL, Koç University *Istanbul, Turkey*
Undergraduate Research Assistant *01/2022 - 03/2022*

- Participated in research project, "IMF Decides, Machine Learns" to achieve a high accuracy prediction of IMF decisions. Assisted in building a large dataset of IMF transcripts.
- Python, Office

KUIS AI Lab, Koç University *Istanbul, Turkey*
Summer Research Intern *06/2021 - 09/2021*

- Research and study on style transfer and image manipulation using GANs. Reimplemented the paper "Swapping Autoencoder for Deep Image Manipulation" by Park et al. (2020).
- Python, PyTorch



ABOUT ME

Focus on neural rendering, 3D reconstruction, generative video models, and data compression.

SKILLS

Python
PyTorch
MATLAB
C/C++
CUDA Programming

LANGUAGES

English • Fluent (TOEFL 2025 115/120)
German • Beginner
Turkish • Native

PROJECT TOPICS

- Neural and generative video/image compression
- Generalizable 3D reconstruction using MVS and Gaussian splatting
- 3D Gaussian splatting representation compression
- Sparse neural 3D reconstruction
- Transformer-based autoregressive DNA sequencing

PUBLICATIONS

- E. Cetin, et al. [Name Redacted for Review Purposes] submitted to CVPR 2026 (under review).
- E. Cetin, M.A. Yilmaz, A.M. Tekalp. "Flexible-Rate Learned Hierarchical Bi-Directional Video Compression," in ICIP, Bordeaux, France, 2022.

PROFESSIONAL EXPERIENCE

Visium SA *Zurich, Switzerland*

Junior Machine Learning Engineer *08/2023 - 02/2024*

- Developed a document storage and retrieval system based on Retrieval-Augmented Generation for a client. Scoped and maturized a production pipeline optimization project.
- Python, PyTorch, Project Scoping, Collaboration, Use-case Development

ART Labs *Istanbul, Turkey*

Computer Vision Intern *07/2021 - 09/2021*

- Designed a 3D garment texture optimizer using multiple still images and 3D model.
- Python, PyTorch, OpenCV

Softtech *Istanbul, Turkey*

Natural Language Processing Intern *06/2021 - 07/2021*

- Assisted building a text classification model and preparing a text corpus for a client bank.
- Python, PyTorch

TEACHING EXPERIENCE

ETH Zurich *Zurich, Switzerland*

Student Teaching Assistant for Large Language Models Lectures *02/2025 - 09/2025*

- Assist in giving tutorial on transformers, arranging assignments, and grading them for the Large Language Models course at ETH Zurich. Around 300 students take the course each year.
- PyTorch, Language Modeling, Deep Learning, Tutoring, Presentation, Public Speaking

ETH Zurich *Zurich, Switzerland*

Student Teaching Assistant for Machine Perception Lectures *02/2024 - 06/2024*

- Prepared and presented tutorials for 417 students in the Machine Perception course. The course introduces a wide-range of Computer Vision applications and is offered to a diverse group of students.
- PyTorch, Computer Vision, Deep Learning, Tutoring, Presentation, Public Speaking

Koç University *Istanbul, Turkey*

Teaching Assistant in ECON 333, Game Theory *09/2020 - 01/2021*

- Teaching assistant for the ”Game Theory & Strategy” lectures. I conducted problem sessions for over 80 students weekly and arranged office hours.

Educational Tutor in ELEC 100, Intro to EEE *02/2019 - 06/2019*

- Tutoring over 90 freshman students in the ”Introduction to EEE” lectures for a support program provided by Koç University.